OLOHIMAI JULIET MICHAEL

Machine Learning Researcher | Trustworthy & Reliable AI | Communication Systems

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GitHub | Google Scholar | LinkedIn | ClimateNetAI Live | ClimateNetAI Repository

RESEARCH PROFILE

Researcher working at the intersection of trustworthy machine learning, distribution shift, generalization, reliability-aware adaptation, and intelligent communication systems. Doctoral work uses longitudinal 5G and atmospheric observations to study how predictive performance changes under temporal and environmental shift, motivating current research toward out-of-distribution (OOD) generalization, transferable representations, and few-shot domain adaptation.

RESEARCH INTERESTS

Out-of-Distribution Generalization | Distribution Shift | Deep & Transferable Representations | Few-Shot Transfer / Domain Adaptation | Robust & Reliable ML | Uncertainty Quantification | Reliability-Aware Learning | ML for Wireless and Cyber-Physical Systems

EDUCATION

PhD, Systems Engineering — African University of Science and Technology (AUST), Abuja, Nigeria | Expected Dec 2026

- Research focus: machine-learning modelling and generalization of 5G signal behaviour under atmospheric, seasonal, and temporal variability.
- Current direction: distribution-shift evaluation, OOD generalization, reliability-aware adaptation, and transfer across environmental/geographical domains.

M.Eng., Information and Communication Engineering — Covenant University, Nigeria | 2012

B.Eng., Information and Communication Engineering — Covenant University, Nigeria | 2008 | Second Class Upper Division

FLAGSHIP RESEARCH & SOFTWARE

ClimateNetAI / RAC-5G — Project Originator, Research Lead & Lead Developer | 2026–Present

Live application | GitHub repository | Zenodo DOI: 10.5281/zenodo.21892444

- Originated and lead the research and software development of a reliability-aware, climate-adaptive ML platform for 5G systems.
- Designed the lifecycle: prediction → uncertainty → monitoring → degradation detection → controlled adaptation → post-adaptation recovery verification.
- Integrated climate-aware RSSI prediction, live atmospheric data, telemetry, rolling error/coverage monitoring, reliability states, and sustained-degradation detection.
- Established a reliability principle central to the project: adaptation does not itself constitute recovery; new post-adaptation evidence is required before reliability is considered restored.
- Conducted controlled functional validation while explicitly separating simulated software validation from independent field-performance evidence.
- Extended ClimateNetAI with RAC-5G and Reliability-Bounded Estimation (RBE), exploring model reliability boundaries, distribution shift, and evidence-based recovery assessment.

Doctoral Research — ML Generalization of 5G Signal Behaviour under Atmospheric Variability

- Conducted empirical ML research using longitudinal 5G and atmospheric observations; evaluated Linear Regression, Decision Tree, Random Forest, XGBoost, LSTM, and GRU.
- Compared conventional and temporally structured validation and identified substantial deterioration under chronological evaluation, motivating subsequent work on distribution shift, robustness, and trustworthy ML.
- Emerging postdoctoral direction: study whether representations learned from source environmental/geographical domains can adapt reliably to unseen domains using only limited target-domain observations.

SELECTED PUBLICATIONS & RESEARCH OUTPUTS

- Michael, O. J. et al. (2026). “When Simpler Models Win: Evaluating Machine Learning Generalization for Climate-Aware 5G Signal Prediction.” Accepted Poster, Women in Machine Learning (WiML) Workshop @ NeurIPS 2026, Atlanta, USA.
- Iruemi, O. J., Ale, F., Momoh, J., Agboola, O., & Ayegba, A. (2025). “Machine Learning-Based Analysis of Weather Influence on 5G Network Signal Propagation in North-Central Nigeria.” Journal of Applied Bioanalysis. DOI: 10.53555/jab.v11i7.1594.
- Iruemi, O. J., Ale, F., Momoh, J., Agboola, O. A., & Ayegba, A. (2024). “Assessment of Seasonal Variation of 5G Signal Strength and Signal Speed over Abuja, North-Central Nigeria.” Journal of Electrical Systems, 20(3), 9347–9355.
- Iruemi, O. J., Ale, F., Momoh, J., Ikpaya, D. I., Agboola, O., & Ayegba, A. “Impact of Some Atmospheric Parameters on 5G Network Signal in Abuja During the Rainy Season Using an Experimental Approach.” Proceedings of the 8th URSI-NG Annual Conference. DOI: 10.2991/978-94-6463-644-4_8.
- Ayegba, A., Iruemi, O. J., et al. (2025). “Enhanced Security Features of 6G Network: A Leap Beyond 5G Network.” American Journal of Technology Advancement, 2(9), 4–11. DOI: 10.31149/ajta.v2i9.2428.
- Iruemi, O. J., et al. (2024). “Applying Cyber Security Engineering to Aerospace Engineering Sector.” International Journal of Advances in Engineering and Management, 6(9), 52–58. DOI: 10.35629/5252-06095258.

SELECTED CONFERENCE RESEARCH, AWARDS & ACADEMIC SERVICE

- AI Predictive Modelling of 5G Signal Behaviour under Atmospheric Variability for National Development — STICOMMS 2026, AUST, Abuja.
- Strengthening Nigeria's Cybersecurity Resilience Through a Critical Infrastructure Vulnerability Repository — ICEGOV 2025, co-author; Best Paper Award.
- Reviewer — Machine Learning for Physical Sciences (ML4PS) 2026.
- Accepted Poster — Women in Machine Learning (WiML) Workshop @ NeurIPS 2026.
- Participant — ACM AI Leadership Summit 2026.
- Certificate of Appreciation — Digital Bridge Institute (DBI), International Girls in ICT Day 2026 Celebration, 23 April 2026; recognized for outstanding contributions as a Guest Speaker. The certificate was presented in conjunction with the International Telecommunication Union (ITU).
- Impact Scholar Reviewer (Volunteer) — Neuromatch; contributed to the Impact Scholar review process.
- Accepted Participant — Radical AI Founders Masterclass, 2026; currently engaging with the programme through its YouTube video sessions.
- Accepted Full Paper — International Conference on the Social Impact of AI (SIAI 2026), Paper #90: "AI for the Connected Many: Climate-Aware Machine Learning for Equitable and Resilient Digital Infrastructure in the Global South."

RESEARCH & ENGINEERING EXPERIENCE

National Space Research and Development Agency (NASRDA), Abuja, Nigeria

Principal Communication Engineer | Feb 2025–Present

- Lead/support systems engineering for satellite communication infrastructure, ground-station systems, technical evaluation, integration, operational reliability, and R&D in satellite, wireless, and intelligent communication systems.

Senior Communication Engineer | Jan 2022–Feb 2025

- Satellite ground-station operations, tracking and mission-control support; communication-systems engineering; technical data management; systems monitoring; and R&D.

Communication Engineer I | Aug 2018–Dec 2021

- Supported satellite ground-station operations, tracking and mission-control activities, communication-system monitoring and maintenance, technical data management, and related research.

Vivid Global Resources — Systems Analyst | Jan 2013–Jul 2018

Covenant University, Ota, Nigeria — Graduate Assistant / Assistant Lecturer | Aug 2009–Nov 2012

- Supported undergraduate teaching and laboratories in Information and Communication Engineering, academic research, and departmental activities.

TECHNICAL SKILLS

Machine Learning: Python, scikit-learn, XGBoost, TensorFlow/Keras, PyTorch; regression, tree-based learning, boosting, LSTM/GRU, model evaluation and validation.

Trustworthy ML: distribution-shift evaluation, temporal validation, uncertainty/prediction intervals, reliability monitoring, degradation detection, adaptation, and recovery verification.

Development & Systems: Streamlit, Git/GitHub, data processing and visualization; satellite ground stations, mission control, wireless communications, 5G systems, network monitoring.

PROFESSIONAL MEMBERSHIPS

COREN Registered Engineer (Information Technology Engineering) | ACM | IEEE | INCOSE | CSTA

SELECTED LEADERSHIP & OUTREACH

African Institute for Innovative Computer Education and Skills Advancement (AICESA) — Advisory Council Member | October 2026–Present.

WechimAI Integrated Engineering & ICT Ltd — President & CEO | AI, automation, digital transformation and technology strategy.

Technology Development for the Girl Child Initiative — Founder & President | STEM outreach and mentorship.

TechWomen INSPIRE Abuja — STEM Mentor / Volunteer Facilitator | Outreach reaching 120+ students.